

# Virtual Site Visits through Gamification for AEC Students: Perspectives on Practice

Sean Pickersgill

Creative, University of South Australia  
Australia  
sean.pickersgill@unisa.edu.au

Damian Madigan

University of South Australia  
Australia  
damian.madigan@unisa.edu.au

Andrew Lymn-Penning

University of South Australia  
Australia  
andrew.lymn-penning@unisa.edu.au

Darcy Holmes

University of South Australia  
Australia  
darcynh@live.com.au

## Abstract

This paper covers the evolution of a unique learning and teaching solution devised for introductory construction education for tertiary architecture and construction students. Since 2015, a simplified version of the VLE model was used to supplement construction curriculum taught in a traditional manner. In 2024, the new and current version of this VLE model *SiteSeer* was introduced in response to user testing and industry feedback. Providing a far more comprehensive and detailed immersive experience, the project continued to strive for authenticity as a substitute for real-world construction site visits. This paper reports on the premise for the project as a gamified platform that continues to address a core challenge of information gathering and authentic learning within a tertiary curriculum.

## CCS Concepts

• **Applied computing** → Education; Interactive Learning Environments; Education.

## Keywords

Virtual Learning Environments, Architecture, Authenticity, Gamification, Construction

## ACM Reference Format:

Sean Pickersgill, Andrew Lymn-Penning, Damian Madigan, and Darcy Holmes. 2024. Virtual Site Visits through Gamification for AEC Students: Perspectives on Practice. In *SIGGRAPH Asia 2024 Educator's Forum (SA Educator's Forum '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3680533.3697062>

## 1 Introduction: onsite/siteseer

The *OnSite/Site Seer* Project at the University of South Australia has involved over 1000 students over nine years as part of a first-year introductory course on Australian domestic construction practices. This course, like many across Australian architecture and building

programs, focusses on familiarizing students with conventional domestic construction methods, assuming no prior knowledge. The original *OnSite* project employed models of a volume-builder house, demonstrating detailed characteristics of the construction process from site setout to the timber framing stage. [Pickersgill et al., 2020] In 2024, the authors revised this approach, developing a new digital model, *SiteSeer*, that incorporated significantly larger curriculum content. The new and current model contains 73 distinct phases of the construction process, including detailed location-specific information regarding material and constructional standards. This paper will outline the pedagogical approach to a virtualised site visit experience, report on the manner of engagement by students and reflect on the general challenges associated with socialising complex BIM (Building Information Modelling) content for novitiate students of architecture and construction.

## 1.1 Methodology

The *SiteSeer* project addressed two primary questions: how to meaningfully expose students to domestic construction processes and what alternatives exist if physical site visits are unfeasible. The original *OnSite* project proposed that Virtual Learning Environments (VLEs) could simulate real-life learning experiences while meeting the pedagogical goals associated with authentic learning. Despite the extensive literature on the potential for VLEs, gamification, constructivist learning, and Learning Management Systems (LMS), there is limited analysis on projects that fully integrate these digital education tools for architecture and construction in a long-range test setting. [Pickersgill et al., 2020]

For example, Maghool, Moeini, and Arefazar's *LADUVR* project highlighted the potential of VLEs in architectural education, but it lacked focus on site visits and LMS integration. [Maghool et al., 2018] Projects often prioritize gamification over visual authenticity, reflecting the digital game theory debate between ludological (instrumental play) and narrative (immersive) approaches. *OnSite* aimed to blend complex software use with the creation of visually authentic VLEs, integrate these with LMS, and apply them to large data sets. The challenge for any VLE is whether they can effectively substitute the experiences of site information without the logistical and health and safety challenges of real-world visits. In addition, the degree of para-informational content within a site-visit VLE should leverage the software capabilities of a Graphic User Interface (GUI) to support and integrate learning objectives outlined within

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

*SA Educator's Forum '24*, December 03–06, 2024, Tokyo, Japan

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1136-7/24/12

<https://doi.org/10.1145/3680533.3697062>



**Figure 1: (left) Screenshot of Original *OnSite* Model (2015-2022) (Authors); (right) Detail of original *OnSite* Model (Authors)**

the parallel LMS modules. For *OnSite/SiteSeer* this involved creating virtual environments that mimic construction site visits, while incorporating relevant contextual information within the GUI that is specific to the external LMS content.

Overall, the *OnSite/SiteSeer* project explores whether virtual learning environments (VLEs) can effectively replace real-world experiences in teaching domestic construction principles. It aims to develop an immersive, information-rich VLE that integrates seamlessly with Learning Management Systems (LMS) and includes assessment modules. The project focuses on creating virtual environments that emulate construction site visits, using accessible software for distribution across various digital platforms. The goal is to provide authentic learning experiences while ensuring students engage with and understand the pedagogical content.

## 1.2 Introduction to Construction

Introducing students to domestic construction principles often involves site visits to observe real-world methods, echoing the apprenticeship model where trade skills are learned alongside skilled craftspeople. While apprenticeships are still prevalent in vocational training (VET) and Technical and Further Education (TAFE), tertiary architecture and building degrees face challenges in providing this first-hand experience due to time constraints. Therefore, these degrees adopt a compromise, offering foundational knowledge of construction principles without extensive hands-on training. For architecture students, this foundational knowledge is crucial as it supports the development of creative skills and innovative construction methods. First-year construction courses focus on familiarizing students with conventional practices, without expecting innovation at this stage. These courses also aim to introduce students to various media used to document and communicate construction knowledge. In professional contexts, construction drawings are key in communicating design details such as scale, method, and material to tradespeople. However, tertiary courses do not assume prior familiarity with these conventions.

In 2015, the UniSA team developed a Virtual Learning Environment (VLE) called *OnSite* to immerse students in a digital construction site, capturing five stages of the construction process: site setout, slab, single-storey wall framing, double-storey wall framing, and roof framing. Using Unreal Engine 4 (UE4), the original *OnSite* provided a realistic digital environment where students can navigate a construction site from a first-person perspective similar to commercial video games. (Figure 1)

Although *OnSite* was not a full "sand box" environment, it allowed students to freely explore a construction site, observing different stages of the building process. The emphasis was upon a mode of informed exploration that encouraged students to observe the state of the model and perform an assessment of its relative completion. In the *OnSite* version no GUI was employed to provide additional information, so students relied on a 'raw' experience of traversing the site and noting its compliance with standards of construction detailed in external educational information, including construction drawings. This relationship reflected a belief that the legitimacy of the VLE came from the specific *lack* of paratextual information in the game experience. Subsequent analysis of student responses and a revision of the pedagogical aims of the teaching staff indicated that a significant potential of game environments generally and VLEs in particular was the opportunity to co-locate significant commentary on the construction process adjacent to the immersive experience of the VLE.

## 2 A New Model: SiteSeer

Thus, a core goal of these courses is to teach students the relationship between construction drawings and their built reality, emphasizing the translation of two-dimensional plans into three-dimensional structures. This foundational understanding is essential for appreciating how construction occurs and forms the basis for future innovation. Reflecting on the limited scope of the original *OnSite* Project, the *SiteSeer* team created a new, more comprehensive building model that included every phase of the construction process of a small, double-storey studio/residence. The new structure was a mixture of timber frame and brick veneer construction, demonstrating a broad scope of construction instances that a first-year student would be expected to understand. (Figure 2)

The *SiteSeer* Project was integrated into a tertiary course focused on introducing students to domestic construction principles and relevant skills. The course includes assessment points designed to teach various construction concepts and documentation methods. The first assessment, a vertical sculpture, required students to construct the highest open frame structure from a single sheet of balsa, introducing them to the idea of structure. The second assessment involved the design and documentation of a small pavilion, which helped students understand the relationship between two-dimensional orthographic drawings and a small building. The third assessment, employing the virtual learning environment and



Figure 2: Screenshot of SiteSeer Model, GUI, (Authors)

LMS-based quizzes, provided the clearest emulation of a three-dimensional understanding of how construction takes place, bridging the gap between drawings and built reality that would normally be achieved by visiting and tracking a real-world construction site.

## 2.1 Site Challenges

The *OnSite* project specifically addressed the challenge of replicating the traditional construction site visit within a virtual learning environment (VLE). By comparing construction drawings with a virtual three-dimensional environment, students could gain practical insights into construction processes, which would typically be achieved through physical site visits. This approach was necessary due to several challenges associated with traditional site visits. Ensuring consistency of experience among all students was difficult without organizing site visits for the entire class. Coordination issues arose because it was challenging to find construction sites that matched the curriculum timeline, and there was no guarantee that the site's construction schedule would align with the students' learning progress. The organizational complexity of arranging access to construction sites also required significant liaison with builders, which added to the workload of academic staff. Furthermore, safety and liability concerns (OHSW) associated with having students on construction sites posed significant risks. High absentee rates were observed when site visits were scheduled outside of regular teaching hours, indicating students' reluctance to attend these visits. Additionally, travel distances to appropriate sites were often impractical, making attendance difficult for students who relied on public transport or had long commuting times. This situation has been noted across many institutions, not only the University of South Australia, and remains a crucial challenge for effective education. [Eiris Pereira and Gheisari, 2017]

## 2.2 A Compromise Model

To address these challenges, a compromise model was previously implemented, where students were responsible for sourcing, contacting, and visiting a construction site independently to maintain a visual diary. However, this approach led to a diminished emphasis

on the diary's importance in assessments and reduced student effort in completing the task. The *OnSite* project aimed to overcome these limitations by providing a consistent, coordinated, and safe virtual learning experience that effectively replaced the traditional site visit, ensuring that students gained essential construction knowledge in a controlled and accessible manner.

## 2.3 Game Engines and Gamification

At this point it is worth clarifying that the pedagogical emphasis within *OnSite/SiteSeer* evolved from a mode of engagement that was generally observational (*OnSite*) to one that encouraged students to undertake a dialectical relationship between the environment and the paratextual information within the GUI (*SiteSeer*). Significantly this classifies the project as primarily a hybrid learning model rather than game-based learning. While it uses game engine software to create realistic environments, it focuses on didactic learning, encouraging students to explore and check the site's compliance with other media rather than forms of play and achievement-based progress associated with game-based learning. It emphasizes observational skills, requiring students to recall anomalies in the built work for online quizzes. Although game-based learning generally involves achieving tasks within simulated environments, both *OnSite* and *SiteSeer* prioritize visual fidelity and realistic site inspection stages as indicators of authenticity of experience, rather than the abstractions in gamification learning models that signify that the game-task rather than the environment has priority. As we will see later, the strength of *SiteSeer's* approach is the commitment to a high degree of ontological fidelity to a simulated environment. (Figure 3)

## 2.4 Game-Based versus Gamification

As a comparison, Carina González and Francisco Blanco's use of the game *Neverwinter Nights* to integrate secondary school mathematics with Moodle, the progress-tracking LMS, illustrates the primacy of the task over the environment, despite the contrived employment of the game. In this quest-based environment, students advance through the game by completing math tasks, combining

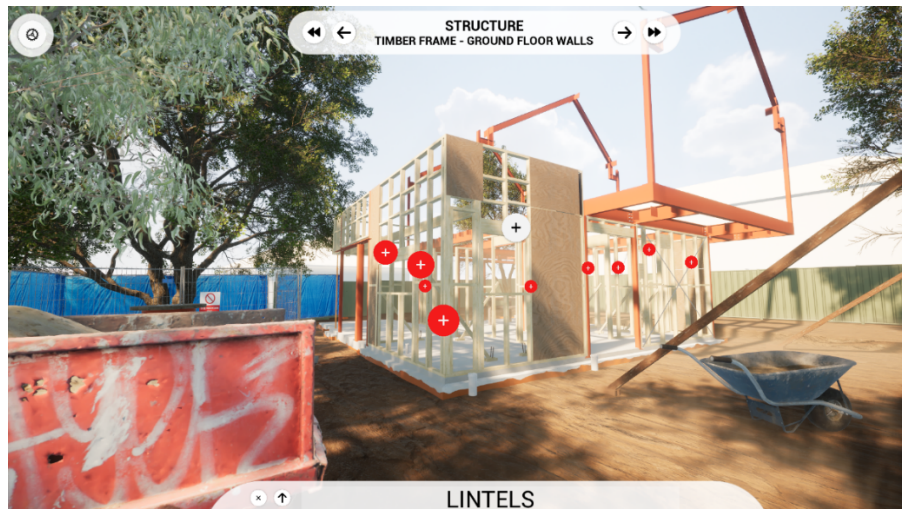


Figure 3: Screenshot of *SiteSeer* Model, Preliminary Structure Phase, 2024 (Authors)

entertainment with skill improvement. Moodle tracks progress, ensuring educational goals are met. González and Blanco's model successfully promotes collaborative learning and social interaction, despite its low-fidelity, fantasy setting. For their purposes, the simplistic graphics were adequate and maintained the game's entertainment value. [González and Blanco, 2008] In contrast, because *OnSite* and *SiteSeer* emphasize realism and immersion as core aspects of their pedagogical approach the structure of the experience relies on a strong relationship between simulation and authenticity in terms of the gamification components of the project. Though *OnSite* isn't a game in the traditional sense and doesn't require social constructivist behavior, it aligns with game-based learning through its use of simulation and iterative experience. Literature from journals like *Simulation and Gaming* reveals key distinctions in game-based learning. [Davies and Krame, 2024][González and Blanco, 2008][Kulkarni et al., 2022][Petko et al., 2020] 'Simulation' can mean either a reiterative game testing multiple outcomes or a high-fidelity replication of real scenarios. In the former, realism is secondary to gameplay, while in the latter, it is paramount. For instance, a chess game's strategy remains unaffected by lifelike pieces, and an elaborate narrative may detract from gameplay quality.

## 2.5 User Responses

In course response surveys students found the original *OnSite*'s synthetic environment a convincing approximation of a typical construction site.[Pickersgill et al., 2020] The subsequent evolution of the project, *SiteSeer* achieved an even greater level of simulated realism, based on both the fidelity of the digital modelling to a construction site as well as the anecdotally important entourage elements (bins, vegetation, tools, etc.) that provided an additional degree of indexical realism to the experience. The balance between realism and educational value highlights *SiteSeer*'s unique approach within the broader context of game-based learning. These responses reflect a similar attitude to the question of fidelity expressed by Davies and Krame in their study of simulation-based learning exercises. [Davies and Krame, 2024] Where Davies and Krame were

focused on the immersive effects of time-critical scenarios, the *SiteSeer* project placed the emphasis on deep fidelity to construction processes.

The current project, *SiteSeer*, further extends the gamification aspects of the overall project. The inclusion of a GUI overlay that included a description of the construction phase, floating markers that indicated information 'hot spots', and the presence of comprehensive text and image-based information provided enhanced levels of information which, while ontologically at odds with immersion, satisfied the greater challenge of educative authenticity. Students understood the simulated environment was aimed at consistency with reality while also tacitly acknowledging that *SiteSeer* was also a focused and artificial experience that required the students to examine or be conscious of it as a learning environment. For the reason that it is possible to undertake a site visit and the information exchange to be merely part of the continuum of experience, *SiteSeer* attempted to make its 'realism' a confirmation of the legitimacy of the information it contained.

## 3 Context and Methods

The *OnSite/SiteSeer* project was originally agreed to by teaching staff on the following conditions: that it demonstrated ease of use, it supported teaching staff and benefitted their construction specialization, it required minimal technical support, and the learning curve for student expertise within the environment was short. In answer to these challenges the following occurred:

- Students successfully downloaded and ran the VLE software for *OnSite/SiteSeer* across a number of personal learning devices, as well as devices available in UniSA learning centres.
- Minimal staff intervention was necessary to solve technical issues regarding the access and use of the *OnSite* VLEs. Forum questions were minimal over the nine-year period.
- Teaching staff who were not originally involved in the technical development of the *OnSite* material in 2015 subsequently gained familiarity with the process, permitting them to rely

on the information content contained within both the *OnSite* and *SiteSeer* projects.

- Construction staff within the Architecture Program at the University of South Australia subsequently contributed to the development of the *SiteSeer* project, significantly extending the scope and depth of pedagogical content and further embedding it within the introductory construction stream.
- Significant anecdotal evidence emerged of cooperative learning behaviour between students based on the shared experience of the *SiteSeer* project, successfully highlighting the fundamentally constructivist mode of learning within the model.

### 3.1 Immersivity

In Chris Dede's original research into immersive teaching environments, the question of immersivity is addressed as a pillar of engagement and learning. Dede was clear that synthetic approaches to place-making in serious games provided a level of engagement that enhanced learning. [Dede, 2009] With respect to the developments in content and immersivity demonstrated in the transition from *OnSite* to *SiteSeer* four key questions emerge:

- How does instructional design for immersive environments vary by subject or learner characteristics, and when is full sensory immersion essential?
- Can successes in virtual identities enhance self-efficacy and real-world educational progress?
- What is the best blend of situated learning across real, augmented, and virtual settings for effective transfer?
- How can insights from immersive environments inform face-to-face pedagogical strategies?

*OnSite/SiteSeer* aligns with these questions by demonstrating the potential of simulated environments for enhancing learning in architecture and construction. By providing authentic learning experiences that emulate real construction conditions, *OnSite* and *SiteSeer* prove, we argue, that significant innovation in this learning domain is both possible and achievable.

### 3.2 Authenticity

Importantly, the form of educational experience associated with gamified environments such as the *OnSite/SiteSeer* VLE can be connected to best-practice in educational experience. Our Project has been guided by the conditions set out by Jan Herrington and Jenni Parker in their study of the conditions for authentic learning within the digital sphere. [Herrington and Parker, 2013] Their study and subsequent literature on immersive software, gamification and authentic learning such as Fischer and Barabach [Fischer and Barabach, 2020] highlights several necessary conditions. These are defined by Jan Herrington and Jenni Parker as (i) Authentic context; (ii) Authentic tasks and activities; (iii) Access to expert performances; (iv) Multiple perspectives; (v) Collaboration; (vi) Reflection; (vii) Articulation; (viii) Scaffolding; (ix) Authentic assessment.

*SiteSeer*, we argue, meets these conditions by offering a consistent, accessible site for all students, appropriate for learning objectives in construction content and sequencing. A dedicated Moodle

Learning Plan ensures students engage with the material in a specific sequence, supporting self-directed learning. This approach underscores the effectiveness of immersive digital environments in providing authentic and impactful learning experiences. Analyzing the effectiveness of *SiteSeer* using Herrington and Parker's principles we found:

- **Authentic Context:** The fidelity to a real-world example and incorporation into summative assessment underscored the project's authenticity. The course focused on familiarizing students with specific knowledge, validating the authenticity of *SiteSeer*.
- **Authentic Tasks and Activities:** The course's relevance to preparing practice-ready students made this implicit. Despite differing applications of knowledge between architecture and building students, core knowledge was recognized and reinforced by teaching staff.
- **Access to Expert Performances:** The paratextual content contained within the text-based supplementary material within the GUI provided an immediate, coherent and sequential description of the significance of material and constructional choices represented within the digital environment.
- **Multiple Perspectives:** Students could complete tasks at home or in university computer labs, fostering co-learning. Official forums and unofficial messaging platforms facilitated discussions. Concerns about student collusion were mitigated by encouraging discussions and multiple quiz attempts to reinforce material.
- **Collaboration:** Mutual discussion and collaborative effort were recognized as crucial. With five quizzes of ten multiple-choice questions each, students had numerous discussion points, reinforcing the material through collaboration.
- **Reflection:** Course evaluation instruments included feedback on *SiteSeer*, with positive support noted. However, it was also mentioned that a real-life site visit would have been valuable.
- **Articulation:** Course materials supported exploration of the VLE, allowing students to recognize the core purpose of *SiteSeer* through lecture materials, exemplar material, and differing course documentation media.
- **Scaffolding:** *SiteSeer* was part of a multi-modal construction education approach, simulating authentic site conditions. The project aimed to produce a digital model accurate in scale, material, and construction method, preparing students for more complex construction environments.
- **Authentic Assessment:** Assessment focused on reviewing the site and recognizing compliance with industry standards (e.g., correct footings, timber studwork). This assessment reflected future professional tasks, ensuring students could identify correct standards for a safe, well-organized construction site. The assessment was thus appropriately authentic.

## 4 Summary

The evolution of the *OnSite* project to the *SiteSeer* version was the outcome of nine years of student testing in the management of introductory construction education. While the original model had

value in introducing a selected number of phases of a domestic-scaled construction project effectively integrated these principles, providing a realistic and immersive learning experience that reinforced core construction knowledge and skills.

## References

- A. J. Davies, and G. Krame. 2024. Measuring the Level of Fidelity Required for Transfer of Learning in Simulation-Based Learning Exercises for Novice and Experienced Practitioners. *Simulation & Gaming*, 55(4), 685-715. <https://doi.org/10.1177/10468781241241548>
- C. Dede. 2009. Immersive Interfaces for Engagement and Learning. *Science (American Association for the Advancement of Science)*, 323(5910), 66–69. <https://doi.org/10.1126/science.1167311>
- R. Eiris Pereira, and M. Gheisari. 2017. Site Visit Application in Construction Education: A Descriptive Study of Faculty Members. *International Journal of Construction Education and Research*, 15(2), 83–99. <https://doi.org/10.1080/15578771.2017.1375050>
- Silke Fischer, and Antje Barabasch. 2020. Gamification: A Novel Didactical Approach for 21st Century Learning. In *Vocational Education and Training in the Age of Digitization* (1st ed., pp. 89-). Verlag Barbara Budrich. <https://doi.org/10.2307/j.ctv18dvv1c.8>
- C. S. González, and F. Blanco. 2008. Integrating an educational 3D game in Moodle. *Simulation & Gaming*, 39(3), 399-413. <https://doi.org/10.1177/1046878108319585>
- J. Herrington, and J. Parker. 2013. Emerging technologies as cognitive tools for authentic learning. *British Journal of Educational Technology*, 44(4), 607–615. <https://doi.org/10.1111/bjet.12048>
- B. Kulkarni, R. Banerjee, and R. Raghunathan. 2022. Why Students Should Be Taught Differently: Learner Characteristics, Learning Styles and Simulation Performance. *Simulation & Gaming*, 53(1), 56-74. <https://doi.org/10.1177/10468781211065809>
- S. A. H. Maghool, S. H. I. Moeini, and Y. Arefazar. 2018. An educational application based on virtual reality technology for learning architectural details: challenges and benefits. *Archnet-IJAR: International Journal of Architectural Research*, 12(3), 246.
- D. Petko, R. Schmid, and A. Cantieni. 2020. Pacing in Serious Games: Exploring the Effects of Presentation Speed on Cognitive Load, Engagement and Learning Gains. *Simulation & Gaming*, 51(2), 258-279. <https://doi.org/10.1177/1046878120902502>
- R. S. Pickersgill, R. Rameezdeen, and J. Harvey. 2020. OnSite: the virtual site visit as an environment for construction learning. In *OnSite: the virtual site visit as an environment for construction learning*. IGI Global. <https://doi.org/10.4018/978-1-5225-8452-0.ch009>